

数学物理方法作业

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第十八章

4. 将 f 按本征函数展开:

$$f = \sum_k A_k \Phi_k$$

原问题化为:

$$\begin{aligned} \nabla^2 u + \sum_k A_k \Phi_k &= 0 \\ u|_{\Sigma} &= 0 \end{aligned}$$

另一方面有:

$$\begin{aligned} \sum_k \frac{A_k}{\lambda_k} (\nabla^2 \Phi_k + \lambda_k \Phi_k) &= \nabla^2 \left(\sum_k \frac{A_k}{\lambda_k} \Phi_k \right) + \sum_k A_k \Phi_k = 0 \\ \sum_k \frac{A_k}{\lambda_k} \Phi_k \Big|_{\Sigma} &= 0 \end{aligned}$$

对比得原问题的解为:

$$u = \sum_k \frac{A_k}{\lambda_k} \Phi_k$$

第十九章

1. 对 t 做 Laplace 变换:

$$\mathcal{L}\{u(x, t)\} = U(x, p), \quad \mathcal{L}\left\{\frac{\partial u}{\partial t}\right\} = pU(x, p), \quad \mathcal{L}\left\{\frac{\partial^2 u}{\partial x^2}\right\} = \frac{d^2 U}{dx^2}$$

方程与边界条件化为:

$$\frac{d^2 U}{dx^2} - \frac{p}{\kappa} U = 0, \quad U(x, p)|_{x=0} = \frac{u_0}{p}$$

由方程解出:

$$U(x, p) = Ae^{\sqrt{p/\kappa}x} + Be^{-\sqrt{p/\kappa}x}$$

代入边界条件得:

$$U(x, p) = \frac{u_0}{p} e^{-\sqrt{p/\kappa}x}$$

反演后得:

$$u(x, t) = u_0 \operatorname{erfc} \frac{x}{2\sqrt{\kappa t}}$$

3. 对 t 做 Laplace 变换:

$$\mathcal{L}\{u(x, t)\} = U(x, p), \quad \mathcal{L}\left\{\frac{\partial u}{\partial t}\right\} = pU(x, p), \quad \mathcal{L}\left\{\frac{\partial^2 u}{\partial x^2}\right\} = \frac{d^2 U}{dx^2}$$

方程与边界条件化为:

$$\frac{d^2 U}{dx^2} - \frac{p}{\kappa} U = 0, \quad U(x, p)|_{x=0} = \frac{A}{p + \alpha^2 \kappa}, \quad U(x, p)|_{x=l} = \frac{B}{p + \beta^2 \kappa}$$

由方程解出:

$$U(x, p) = Ce^{\sqrt{p/\kappa}x} + De^{-\sqrt{p/\kappa}x}$$

代入边界条件得:

$$\begin{aligned} U(x, p) &= \frac{e^{-\sqrt{p/\kappa}l}}{e^{\sqrt{p/\kappa}l} - e^{-\sqrt{p/\kappa}l}} \left[\frac{Be^{\sqrt{p/\kappa}l}}{p + \beta^2 \kappa} - \frac{A}{p + \alpha^2 \kappa} \right] e^{\sqrt{p/\kappa}x} \\ &\quad + \frac{1}{e^{\sqrt{p/\kappa}l} - e^{-\sqrt{p/\kappa}l}} \left[\frac{Ae^{\sqrt{p/\kappa}l}}{p + \alpha^2 \kappa} - \frac{B}{p + \beta^2 \kappa} \right] e^{-\sqrt{p/\kappa}x} \\ &= \frac{A \sinh \sqrt{p/\kappa}(l-x)}{(p + \alpha^2 \kappa) \sinh \sqrt{p/\kappa}l} + \frac{B \sinh \sqrt{p/\kappa}x}{(p + \beta^2 \kappa) \sinh \sqrt{p/\kappa}l} \end{aligned}$$

反演得:

$$u(x, t) = \frac{1}{2\pi i} \int_{s-i\infty}^{s+i\infty} U(x, p) e^{pt} dp = \sum_{n=1}^{\infty} \operatorname{res}(P_n)$$

其中 $p = P_n$ 为 $U(x, p)e^{pt}$ 在全复平面内的孤立奇点。

为方便计算留数, 令:

$$V(x, p) = \frac{A \sinh \sqrt{p/\kappa}(l-x)}{(p + \alpha^2 \kappa) \sinh \sqrt{p/\kappa}l} e^{pt}, \quad W(x, p) = \frac{B \sinh \sqrt{p/\kappa}x}{(p + \beta^2 \kappa) \sinh \sqrt{p/\kappa}l} e^{pt}$$

则反演公式变为:

$$u(x, t) = \sum_{n=1}^{\infty} \text{res}(p_n) + \sum_{n=1}^{\infty} \text{res}(p'_n)$$

其中 p_n 为 $V(x, p)$ 在全复平面内的孤立奇点, p'_n 为 $W(x, p)$ 在全复平面内的孤立奇点。则:

$$\begin{aligned} V(x, p) &= \frac{P(x, p)}{Q(x, p)}, & \frac{\partial Q(x, p)}{\partial p} &= \sinh \sqrt{\frac{p}{\kappa}} l + \frac{(p + \alpha^2 \kappa) l}{2\sqrt{p\kappa}} \cosh \sqrt{\frac{p}{\kappa}} l \\ W(x, p) &= \frac{P'(x, p)}{Q'(x, p)}, & \frac{\partial Q'(x, p)}{\partial p} &= \sinh \sqrt{\frac{p}{\kappa}} l + \frac{(p + \beta^2 \kappa) l}{2\sqrt{p\kappa}} \cosh \sqrt{\frac{p}{\kappa}} l \end{aligned}$$

且:

$$\text{res}(p_n) = \frac{P(x, p_n)}{\left. \frac{\partial Q(x, p)}{\partial p} \right|_{p=p_n}}, \quad \text{res}(p'_n) = \frac{P'(x, p'_n)}{\left. \frac{\partial Q'(x, p)}{\partial p} \right|_{p=p'_n}}$$

由 $p + \alpha^2 \kappa = 0$ 及 $p + \beta^2 \kappa = 0$ 可以得到两个特殊奇点:

$$p_0 = -\alpha^2 \kappa, \quad p'_0 = -\beta^2 \kappa$$

其留数为:

$$\begin{aligned} \text{res}(p_0) &= \frac{A \sinh \sqrt{p/\kappa} (l-x)}{\sinh \sqrt{p/\kappa} l} e^{-\alpha^2 \kappa t} = \frac{A \sin \alpha (l-x)}{\sin \alpha l} e^{-\alpha^2 \kappa t} \\ \text{res}(p'_0) &= \frac{B \sin \sqrt{p/\kappa} x}{\sin \sqrt{p/\kappa} l} e^{-\beta^2 \kappa t} = \frac{B \sin \beta x}{\sin \beta l} e^{-\beta^2 \kappa t} \end{aligned}$$

由 $\sinh \sqrt{\frac{p}{\kappa}} l = 0$ 得: (注意 $p = 0$ 为孤立奇点)

$$\sqrt{\frac{p}{\kappa}} l = n\pi i, \quad p_n = p'_n = -\frac{n^2 \pi^2 \kappa}{l^2}, \quad n \in \mathbb{N}^+$$

其留数为:

$$\begin{aligned} \text{res}(p_n) &= \frac{A \sinh(\frac{n\pi i}{l})(l-x)}{\frac{(p + \alpha^2 \kappa) l}{2\sqrt{p\kappa}} \cosh(\frac{n\pi i}{l}) l} e^{-\left(\frac{n\pi}{l}\right)^2 \kappa t} \\ &= 2n\pi \frac{A \sin(\frac{n\pi}{l})(l-x)}{(-1)^n [(\alpha l)^2 - (n\pi)^2]} e^{-\left(\frac{n\pi}{l}\right)^2 \kappa t} \\ &= 2n\pi \frac{A \sin(\frac{n\pi}{l}) x}{(\alpha l)^2 - (n\pi)^2} e^{-\left(\frac{n\pi}{l}\right)^2 \kappa t} \end{aligned}$$

$$\begin{aligned} \text{res}(p'_n) &= \frac{B \sinh(\frac{n\pi i}{l}) x}{\frac{(p + \beta^2 \kappa) l}{2\sqrt{p\kappa}} \cosh(\frac{n\pi i}{l}) l} e^{-\left(\frac{n\pi}{l}\right)^2 \kappa t} \\ &= 2in\pi \frac{B \sin(\frac{n\pi}{l}) x}{(-1)^n [(\beta l)^2 - (n\pi)^2]} e^{-\left(\frac{n\pi}{l}\right)^2 \kappa t} \end{aligned}$$

$$= -2n\pi(-1)^n \frac{B \sin(\frac{n\pi}{l})x}{(\beta l)^2 - (n\pi)^2} e^{-(\frac{n\pi}{l})^2 \kappa t}$$

则:

$$\begin{aligned} u(x, t) &= \frac{A}{\sin \alpha l} \sin \alpha(l-x) e^{-\alpha^2 \kappa t} + \frac{B}{\sin \beta l} \sin \beta x e^{-\beta^2 \kappa t} \\ &+ \sum_{n=1}^{\infty} 2n\pi \left[\frac{A}{(\alpha l)^2 - (n\pi)^2} - \frac{(-1)^n B}{(\beta l)^2 - (n\pi)^2} \right] \left(\sin \frac{n\pi}{l} x \right) e^{-(\frac{n\pi}{l})^2 \kappa t} \end{aligned}$$

5. 定义:

$$\mathcal{F}^2 f(m, k) = \frac{1}{2\pi} \iint_{\mathbb{R}^2} f(x, y) e^{-i(mx+ky)} dx dy$$

其逆变换为:

$$\mathcal{F}^{-2} f(m, k) = \frac{1}{2\pi} \iint_{\mathbb{R}^2} f(x, y) e^{i(mx+ky)} dx dy$$

$$\mathcal{F}^2 u(x, y, t) = U(m, k, t), \quad \mathcal{F}^2 \left\{ \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right\} = -(m^2 + k^2) U(m, k, t)$$

$$\mathcal{F}^2 \Phi(x, y) = \Phi(m, k), \quad \mathcal{F}^2 \psi(x, y) = \Psi(m, k)$$

则原问题化为:

$$\frac{d^2 U}{dt^2} + a^2 (k^2 + m^2) U = 0, \quad U|_{t=0} = \Phi(m, k), \quad \left. \frac{dU}{dt} \right|_{t=0} = \Psi(m, k)$$

解得:

$$U(m, k, t) = \Phi(m, k) \cos(a\sqrt{m^2 + k^2}t) + \frac{\Psi(m, k)}{a\sqrt{m^2 + k^2}} \sin(a\sqrt{m^2 + k^2}t)$$

设:

$$h(x, y, t) = \mathcal{F}^{-2} \left[\frac{\sin(a\sqrt{m^2 + k^2}t)}{a\sqrt{m^2 + k^2}} \right]$$

则由 Fourier 变换的卷积定理:

$$U(k, m, t) = V(k, m, t) + W(x, y, t) = \mathcal{F}^2 \psi(x, y) \mathcal{F}^2 h(x, y, t) + \frac{\partial}{\partial t} [\mathcal{F}^2 \Phi(x, y) \mathcal{F}^2 h(x, y, t)]$$

对第一项做反演:

$$v(x, y, t) = \frac{1}{2\pi} \iint_{\mathbb{R}^2} \psi(x', y') \left\{ \frac{1}{2\pi} \iint_{\mathbb{R}^2} \frac{\sin a\lambda t}{a\lambda} e^{i[m(x-x') + k(y-y')] } dmdk \right\} dx' dy'$$

其中 $\lambda = \sqrt{m^2 + k^2}$ 。

换为极坐标得:

$$v(x, y, t) = \frac{1}{2\pi} \iint_{\mathbb{R}^2} \psi(x', y') \left\{ \frac{1}{2\pi} \iint_{\mathbb{R}^2} \frac{\sin a\lambda t}{a\lambda} e^{i\lambda r \cos(\theta - \Phi)} \lambda d\lambda d\theta \right\} dx' dy'$$

其中 $r = \sqrt{(x - x')^2 + (y - y')^2}$ 。

$$\frac{\sin a\lambda t}{a} = \lambda t \cdot \frac{\sin a\lambda t}{a\lambda t} = \lambda t \cdot \sqrt{\frac{\pi}{2a\lambda t}} J_{\frac{1}{2}}(a\lambda t)$$

$$e^{i\lambda r \cos(\theta - \Phi)} = \sum_{n=0}^{\infty} J_n(\lambda r) e^{in(\theta - \Phi)}$$

由以下公式：

$$\int_0^{\infty} t^{1-\mu+\nu} J_{\mu}(at) J_{\nu}(bt) dt = \frac{b^{\nu}(a^2 - b^2)^{\mu-\nu-1}}{2^{\mu-\nu+1} a^{\mu} \Gamma(\mu - \nu)} \eta(a - b)$$

取 $\mu = \frac{1}{2}$, $\nu = 0$, 得：

$$\begin{aligned} & \iint_{\mathbb{R}^2} \frac{\sin a\lambda t}{a\lambda} e^{i\lambda r \cos(\theta - \Phi)} \lambda d\lambda d\theta \\ &= \iint_{\mathbb{R}^2} \lambda t \cdot \sqrt{\frac{\pi}{2a\lambda t}} J_{\frac{1}{2}}(a\lambda t) \sum_{n=0}^{\infty} J_n(\lambda r) e^{in(\theta - \Phi)} d\lambda d\theta \\ &= \int_0^{\infty} 2\pi \cdot \sqrt{\frac{\pi\lambda t}{2a}} J_{\frac{1}{2}}(a\lambda t) J_0(\lambda r) d\lambda \\ &= 2\pi \cdot \sqrt{\frac{\pi t}{2a}} \int_0^{\infty} \lambda^{\frac{1}{2}} J_{\frac{1}{2}}(\lambda at) J_0(\lambda r) d\lambda \\ &= 2\pi \cdot \frac{\eta(at - r)}{a\sqrt{(at)^2 - r^2}} \end{aligned}$$

故：

$$v(x, y, t) = \frac{1}{2\pi a} \iint_{\mathbb{R}^2} \psi(x', y') \frac{\eta(at - r)}{\sqrt{(at)^2 - r^2}} dx' dy' = \frac{1}{2\pi a} \iint_S \frac{\psi(x', y')}{\sqrt{(at)^2 - r^2}} dx' dy'$$

其中 S 为以 (x, y) 为圆心, at 为半径的圆域。

同理可得：

$$w(x, y, t) = \frac{\partial}{\partial t} \left[\frac{1}{2\pi a} \iint_S \frac{\psi(x', y')}{\sqrt{(at)^2 - r^2}} dx' dy' \right]$$

综上：

$$u(x, y, t) = \frac{1}{2\pi a} \left[\iint_S \frac{\psi(x', y')}{\sqrt{(at)^2 - r^2}} dx' dy' + \frac{\partial}{\partial t} \iint_S \frac{\psi(x', y')}{\sqrt{(at)^2 - r^2}} dx' dy' \right]$$

第二十章

1. 由 Green 第二公式, 考虑:

$$\iiint_V (\Phi \nabla^2 \Phi' - \Phi' \nabla^2 \Phi) dV = \iint_\Sigma (\Phi \nabla \Phi' - \Phi' \nabla \Phi) \cdot d\Sigma$$

由静电场的 Coulomb 定理和 Gauss 定理得:

$$\nabla^2 \Phi = -\frac{\rho}{\varepsilon_0}, \quad \nabla^2 \Phi' = -\frac{\rho'}{\varepsilon_0}$$

$$\nabla \Phi = -\mathbf{E} = -\frac{\sigma}{\varepsilon_0} \mathbf{n}, \quad \nabla \Phi' = -\mathbf{E}' = -\frac{\sigma'}{\varepsilon_0} \mathbf{n}$$

代入得:

$$-\frac{1}{\varepsilon_0} \iiint_V (\Phi' \rho - \Phi \rho') dV = -\frac{1}{\varepsilon_0} \iint_\Sigma (\Phi \sigma' - \Phi' \sigma) d\Sigma$$

整理即得:

$$\iiint_V \rho \Phi' dV + \iint_\Sigma \sigma \Phi' d\Sigma = \iiint_V \rho' \Phi dV + \iint_\Sigma \sigma' \Phi d\Sigma$$

原命题得证。